

Linear Algebra – MAT 2610

Section 4.5 (Dimension of a Vector Space)

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Theorem 10

Let $\mathcal{B} = \{b_1, \dots, b_n\}$ be a basis for vector space V . Then any set in V containing more than n vectors must be linearly dependent.



Theorem 10 - Proof

Let $\{u_1, \dots, u_p\}$ be a set of vectors in V with $p > n$. The coordinate vectors $[u_1]_B, \dots, [u_p]_B$ are linearly dependent because there are more vectors than entries in each vector.

Therefore, there are scalars, not all zero, $c_1, \dots, c_p$ such that

$$0 = c_1[u_1]_B + \dots + c_p[u_p]_B$$

But the coordinate mapping is a linear transformation, so

$$0 = [c_1u_1 + \dots + c_pu_p]_B$$



Theorem 10 - Proof

Since any linear transformation transforms the zero vector into the zero vector, we get

$$\underline{0 = c_1 u_1 + \cdots + c_p u_p} \quad \text{since } x \mapsto [x]_B \text{ is 1-1}$$

But not all the coefficients are zero, so the set $\{u_1, \dots, u_p\}$ is linearly dependent



Theorem 11

If a vector space V has a basis of n elements, then every basis of V has exactly n elements



Theorem 11 - Proof

Suppose $\mathcal{B}_1$ with n elements and $\mathcal{B}_2$ with p elements are both bases of V . By Theorem 10, if $p > n$, then the elements of $\mathcal{B}_2$ are linearly dependent, which is impossible since $\mathcal{B}_2$ is a basis. Likewise, if $n > p$, then the elements of $\mathcal{B}_1$ are linearly dependent, which is impossible since $\mathcal{B}_1$ is a basis.

Hence, we must have that $n = p$



Definition

If a vector space V is spanned by a finite set, then V is **finite-dimensional**. The **dimension** of V , written $\dim(V)$, is the number of vectors in a basis for V . The dimension of the zero vector space $\{0\}$ is zero. If V is not spanned by a finite set, then V is said to be **infinite-dimensional**.



Theorem 12

Let H be a subspace of a finite-dimensional vector space V . Any linearly independent set in H can be expanded, if necessary, to a basis for H . Also, H is finite-dimensional and $\dim(H) \leq \dim(V)$



Theorem 12 - Proof

If $H = \{0\}$ then $0 = \dim(H) \leq \dim(V)$

Otherwise, let $S = \{u_1, \dots, u_k\}$ be any linearly independent set in H . If S spans H , then S is a basis for H . Otherwise, there is some u_{k+1} in H that is not in $\text{Span}(S)$. But then $\{u_1, \dots, u_k, u_{k+1}\}$ is linearly independent.

Continue until the set spans H , but the size of this set can never get any larger than $\dim(V)$ by Theorem 10.



Theorem 13

Let V be a p -dimensional vector space with $p \geq 1$

- a. Any linearly independent set of exactly p elements in V is automatically a basis for V .
- b. Any set of exactly p elements that spans V is automatically a basis for V .



Theorem 13 - Proof

By Theorem 12, a linearly independent set S of p elements can be extended to a basis for V . But that basis must contain exactly p elements, since $\dim(V) = p$. So S must already be a basis for V .

Now suppose that S has p elements and spans V . Since V is nonzero, the Spanning Set Theorem implies that a subset S' of S is a basis of V . Since $\dim(V) = p$, then S' must contain p vectors. Hence $S = S'$



Defintion

The rank of a matrix A is the dimension of $col(A)$

Since $row(A) = col(A^T)$, then $\dim(row(A)) = \dim(col(A))$

The nullity of a matrix A is the dimension of $nul(A)$



Theorem 14

If A is an $m \times n$ matrix, then

$$\underline{\text{rank}(A) + \text{nullity}(A)} = n$$

↑
columns



Theorem 14 - Proof

If A is an $m \times n$ matrix, then $\text{rank}(A)$ is the number of pivot columns, and $\text{nullity}(A)$ is the number of free variables, which is the number of non-pivot columns.



Example

1. If A is a 7×9 matrix with a two-dimensional null space, what is the rank of A ?
2. Could a 6×9 matrix have a two-dimensional null space?

1) # columns = 9, so $\text{rank}(A) = 9 - \text{nullity}(A)$
 $= 9 - 2 = 7$

2) No - there can be at most 6 pivot entries
so $\text{rank}(A) \leq 6$. But $\text{rank}(A) + \text{nullity}(A) = 9$
so $\text{rank}(A) \geq 7$ if $\text{nullity}(A) = 2$.



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